Pharmaceutical Science

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Pharmaceutical Science (A10563)

Short Title:	Pharmaceutical Science
Faculty	Science and Computing
Department	Science
Cluster	Pharmaceutical Science
Author	JPOWER
Credits	5

Level: Introductory

Description of Module / Aims

This module gives an insight into the application of analytical science to pharmaceutical products, and the role of pharmacopoeias in the regulation of the safety and quality of pharmaceutical products. The module gives an overview of pharmaceutical production and the different types of drug formulations.

Programmes

SCIE-0002	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	2 / 3 / M
SCIE-0002	BSc in Pharmaceutical Science (WD_SPHSC_D)	2 / 3 / M

Indicative Content

- Role of pharmacopoeias and ICH in pharmaceutical analysis and regulation; contents of typical monographs
- Overview of types of pharmaceutical formulations; detailed description of tablet production and typical tablet contents
- Structure, function and use of excipients in tablet formulations
- Introduction to statistical quality control in the pharmaceutical Industry; application to QC charts for defective products
- Operating curves and sampling in pharmaceutical analysis: single and double sampling plans
- Assay and identity testing of pharmaceuticals by pharmacopoeial methods; physical tests for tablets
- Structure and degradation of typical pharmaceutical products: reaction types and examples; stability studies and shelf-life prediction
- Information sources for pharmaceuticals; use of library and on line resources

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the role of pharmacopoeias in the pharmaceutical industry.
2. Use a Pharmacopoeial monograph to analyse drug formulations.
3. Describe the content and production of a typical tablet formulation.
4. Discuss, with examples, the stability of drug formulations.
5. Describe the application of statistical quality control in pharmaceutical production.
6. Use Pharmacopeial procedures to identify and test pharmaceuticals.

Learning and Teaching Methods

- Lectures.
- Laboratory practicals.
- Seminars.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,3,4,5
Continuous Assessment	50%	
Practical	50%	2,6

Assessment Criteria

<40%: No understanding of the basic concepts of pharmaceutical science.
40%-49%: Able to understand the basics of pharmaceutical science. May have some difficulty with clear descriptions.
50%-59%: Good level of achievement of learning outcomes.
60%-69%: In addition to the above, good critical thinking abilities. Shows a high level of competence and efficiency in relation to learning outcomes.
70%-100%: All previous to excellent level. Be able to describe and discuss aspects of pharmaceutical science and show initiative in approaching problems.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Khar, R. and S. Vyas. _Theory and Practice of Industrial Pharmacy_. 4th ed. London: CBS, 2013.
- Pharmaceutical, S. _British Pharmacopoeia 2015_. London: HMSO, 2015.

Requested Resources

- Room Type: Chemistry Lab